

# Consumer Segmentation and Restaurant Concept Performance

## A Market Research Case Study Using UCI Restaurant Preference Data

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### Abstract

Market segmentation is useful only when it creates groups that can be described, compared, and acted on. This paper analyzes the UCI Restaurant and Consumer Data set as a compact market research case study. The analysis asks whether consumer profile variables identify distinct restaurant consumer segments, whether restaurant concept families with stronger service cues show higher observed satisfaction, and whether venue attributes add explanatory value after accounting for consumer profile and fit variables. The data include 1,161 user-restaurant ratings from 138 consumers, plus consumer profiles, stated cuisine preferences, payment preferences, and restaurant attributes. I used Gower distance and partitioning around medoids to segment mixed categorical profile data, then evaluated satisfaction differences with chi-square tests and a logistic regression model. The segment solution was descriptively useful, but high-satisfaction rates did not differ significantly across segments,  $\chi^2(2) = 2.99$ ,  $p = 0.225$ . Familiar concepts with alcohol service had a higher high-satisfaction rate than low-price no-alcohol concepts, 46% versus 33%,  $\chi^2(1) = 12.10$ ,  $p < 0.001$ . Venue attributes also improved model fit after profile and fit controls, likelihood-ratio  $\chi^2(5) = 13.02$ ,  $p = 0.023$ . The results support using the observed concept scorecards to generate hypotheses for a follow-up concept test, not as final evidence of causal preference drivers.

## 1 Introduction

Segmentation is one of the older ideas in marketing research, but its value depends on how well the segments clarify a decision. Smith (1956) framed segmentation as an alternative to treating a market as homogeneous, arguing that differentiated demand creates openings for differentiated offers. More recent methodological work is more cautious. Segments need to be interpretable, stable enough for the decision at hand, and linked to a realistic action, otherwise the analyst has only created labels (Dolnicar et al., 2018).

Restaurant choice is a useful setting for a portfolio-style market research project because it combines consumer profile variables, stated preferences, product or venue attributes, and satisfaction outcomes. It also resembles many client research problems: the analyst is not only estimating a model, but deciding what a brand, product team, or client service group should learn next. In a richer primary research design, this question might be answered with conjoint analysis or discrete choice experiments. Conjoint methods were developed to estimate how respondents value product attributes and have long been used in consumer research for product and concept evaluation (Green and Srinivasan, 1978, 1990). The present data set is not a conjoint experiment. It is observational preference and rating data, so the appropriate standard is more modest: identify patterns, test whether those patterns are strong enough to merit attention, and state what a better follow-up study should measure.

The project is designed around three hypotheses. The first asks whether consumer profile fields can produce meaningful segments. The second asks whether a specific concept pattern, familiar restaurants with alcohol service, is associated with higher observed satisfaction than low-price no-alcohol concepts. The third asks whether restaurant attributes still add explanatory value after controlling for consumer profile and consumer-restaurant fit variables. Together, these hypotheses create a practical market research workflow: segment the audience, score concepts, and then test whether the descriptive concept story survives a multivariate check.

## 2 Research Questions and Hypotheses

The study answers one main research question: can consumer profile data and restaurant attributes be combined to identify useful restaurant consumer segments and concept-performance signals?

That question is split into three testable hypotheses:

- **H1:** High satisfaction differs across profile-derived consumer segments.
- **H2:** Familiar restaurant concepts with alcohol service have higher high-satisfaction rates than low-price no-alcohol concepts.
- **H3:** Venue attributes add explanatory value after consumer profile and consumer-restaurant fit variables are controlled.

The hypotheses are intentionally limited. H1 concerns observed satisfaction differences across clusters, not whether the clusters are permanent psychological types. H2 concerns an observed concept contrast, not the causal effect of alcohol service. H3 concerns incremental explanatory value in a logistic model, not a deployable prediction system.

## 3 Data

The data source is the Restaurant and Consumer Data set from the UCI Machine Learning Repository (Medellin and Serna, 2011). UCI describes the data as originating from a restaurant recommender-system prototype. The data include 9 files: restaurant payment methods, cuisine categories, hours, parking, restaurant attributes, consumer cuisine preferences, consumer payment preferences, consumer profiles, and user-restaurant ratings. UCI lists 138 consumer instances, 130 restaurants, and 1,161 rating rows, with missing values present in several files (Medellin and Serna, 2011).

The analytical data set joins the following raw files:

- `rating_final.csv`, containing user, restaurant, overall rating, food rating, and service rating.
- `userprofile.csv`, containing demographic and profile variables such as activity, budget, transportation, drink level, interest, and personality.
- `usercuisine.csv` and `chefmozcuisine.csv`, used to determine whether a restaurant matches a user's stated cuisine preferences.
- `userpayment.csv` and `chefmozaccepts.csv`, used to determine whether a restaurant accepts one of a user's stated payment methods.

- **geoplaces2.csv**, containing restaurant attributes such as price, ambience, alcohol service, accessibility, smoking area, and location fields.

The outcome is high satisfaction, defined as an overall rating equal to 2 on the original 0 to 2 rating scale. This binary outcome was chosen because a score of 2 is the clearest available positive rating. It should not be interpreted as loyalty, repeat purchase, or long-term preference. The data are better suited to short-run preference screening than to durable customer lifetime-value analysis.

## 4 Methods

### 4.1 Data Preparation

The analysis was implemented in R using the tidyverse for data cleaning, joins, transformation, summaries, and visualization (Wickham et al., 2019). Missing values represented as question marks or blank strings were standardized before feature engineering. Age was derived from birth year using 2012, the UCI donation year, as the reference point. BMI was calculated for descriptive profiling but was not treated as a main analytical driver, since the restaurant question does not require a health interpretation.

Three consumer-restaurant fit indicators were engineered:

- **Cuisine match:** whether the user’s stated cuisine preferences included at least one cuisine attached to the restaurant.
- **Payment match:** whether the restaurant accepted at least one of the user’s stated payment methods.
- **Price within budget:** whether the restaurant’s price tier was at or below the consumer’s stated budget tier.

Restaurant concept families were built from three attributes that are easy to interpret in a concept-screening setting: price tier, alcohol service, and ambience. For example, one concept family is “High price — Full bar — Familiar.” This is not a substitute for a proper conjoint design, but it creates a practical bridge between raw venue attributes and client-facing concept language.

### 4.2 Segmentation Method

The segmentation used consumer profile variables only, not ratings. This avoids defining segments by the outcome that is later being compared. The profile fields are mostly categorical, so standard Euclidean distance would be inappropriate. I used Gower distance because it was designed to measure similarity over mixed variable types (Gower, 1971). I then applied partitioning around medoids, a k-medoids method that represents each cluster by an observed case rather than by an artificial centroid (Kaufman and Rousseeuw, 1990). Medoid-based clustering is useful here because the data are categorical and survey-like rather than continuous and spherical.

Candidate solutions with 3, 4, and 5 clusters were compared using average silhouette width and minimum segment size. The silhouette statistic measures how close an observation is to its own cluster relative to the next nearest cluster (Rousseeuw, 1987). Although the 4- and 5-cluster solutions had slightly higher average silhouette values, they produced four-person segments. For a market research deliverable, those tiny clusters would be too fragile to profile or score. The selected solution therefore used 3 clusters, balancing silhouette information with a minimum practical segment size. This follows the broader guidance that segmentation is a decision tool, not

just an optimization exercise (Dolnicar et al., 2018). Modern work on k-medoids also emphasizes the usefulness of medoids for arbitrary dissimilarities, including cases using Gower-style distances (Schubert and Rousseeuw, 2021).

Table 1: Cluster solution diagnostics

| Segments | Average silhouette | Minimum segment size |
|----------|--------------------|----------------------|
| 3        | 0.080              | 32                   |
| 4        | 0.105              | 4                    |
| 5        | 0.108              | 4                    |

### 4.3 Statistical Analysis

The hypothesis tests used three related analyses. For H1, I used a Pearson chi-square test comparing high satisfaction across the three profile-derived segments. This tests whether the observed high-rating distribution differs across segments. It does not evaluate whether the clusters are psychologically deep or stable over time.

For H2, I compared two concept families: familiar concepts with alcohol service and low-price no-alcohol concepts. The first category included familiar restaurants with wine/beer or full-bar service. The second category included low-price no-alcohol restaurants. This contrast was chosen before testing because it captures a clear market research question: whether a higher-service familiar concept appears to outperform a low-price no-alcohol concept in this sample.

For H3, I fit nested logistic regression models. Logistic regression is a standard generalized linear model for binary outcomes, and the generalized linear model framework was introduced to handle outcomes whose distribution and mean-variance relationship differ from ordinary linear regression (Nelder and Wedderburn, 1972). Applied logistic regression texts emphasize that odds ratios should be interpreted with attention to model specification, sample size, and the observational nature of the data (Hosmer et al., 2013). The reduced model included consumer-restaurant fit variables and consumer profile controls. The fuller model added venue attributes: price, alcohol service, and ambience. A likelihood-ratio test evaluated whether those venue attributes improved fit.

## 5 Results

### 5.1 Segment Profiles

The three-segment solution produced interpretable consumer groups, even though the sample was narrow and heavily student-based. The groups were labeled based on their strongest profile markers:

- **S1: Quieter mixed students**, the largest group with 63 consumers and 561 ratings.
- **S2: Mobile social students**, a smaller group with 32 consumers and 264 ratings.
- **S3: Tech-focused students**, a group with 43 consumers and 336 ratings.

Table 2: Segment profile summary

| Segment                    | Customers | Ratings | High satisfaction | Social/casual drinker |
|----------------------------|-----------|---------|-------------------|-----------------------|
| S1: Quieter mixed students | 63        | 561     | 39.6%             | 36.5%                 |
| S2: Mobile social students | 32        | 264     | 45.8%             | 90.6%                 |
| S3: Tech-focused students  | 43        | 336     | 42.6%             | 81.4%                 |

### Segment profile markers

Percent of consumers in each segment with selected survey characteristics

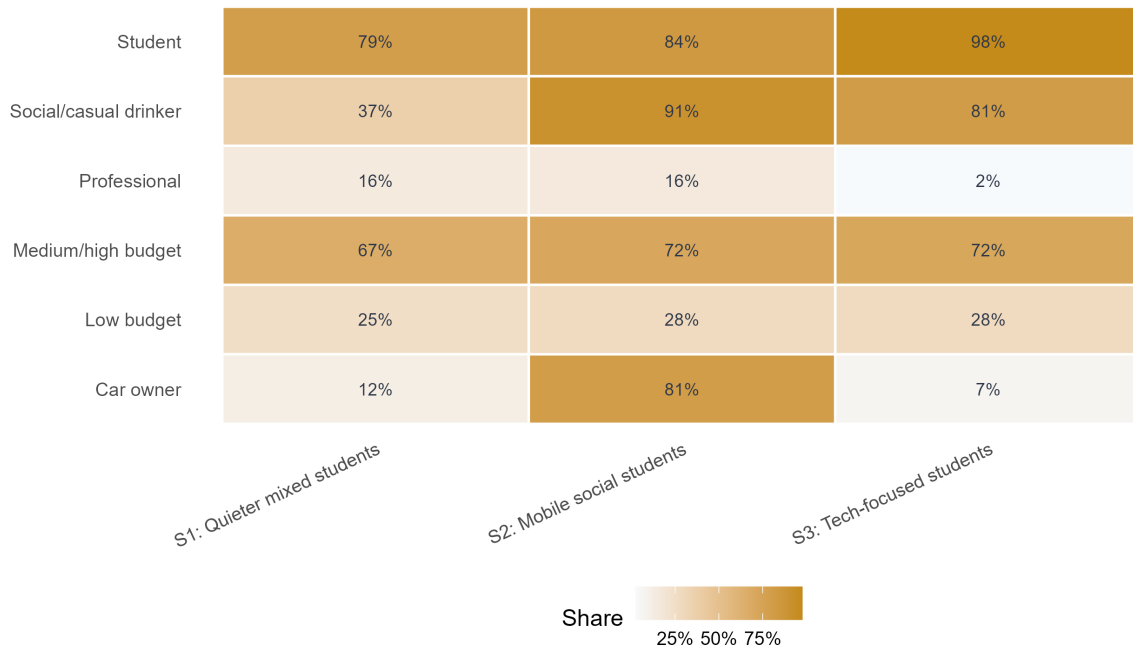


Figure 1: Segment profile markers based on consumer survey fields.

The mobile social segment is the most commercially distinctive profile. In that segment, 91% of consumers are social or casual drinkers and 81% report owning a car. The tech-focused segment is the most student-heavy, with 98% students and a top interest of technology. The quieter mixed segment has lower social-drinker and car-owner shares.

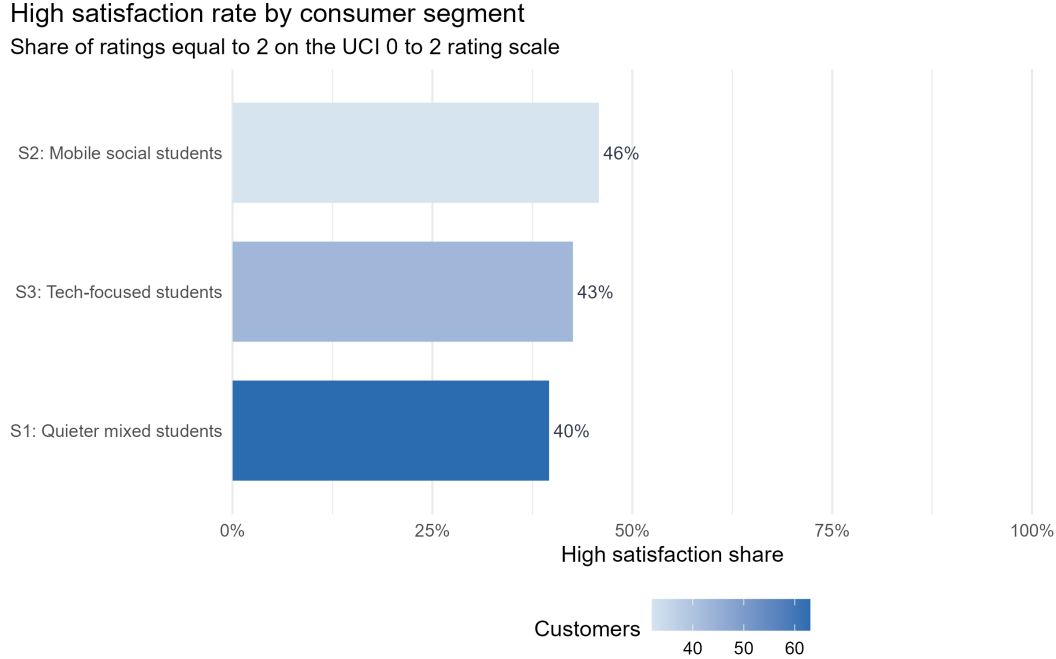


Figure 2: High satisfaction share by consumer segment.

The satisfaction differences are directionally interesting but not statistically strong. Mobile social students have the highest high-satisfaction share at 46%, compared with 43% for tech-focused students and 40% for quieter mixed students. The chi-square test for H1 was not significant,  $\chi^2(2) = 2.99$ ,  $p = 0.225$ . I therefore do not reject the null hypothesis of equal high-satisfaction rates across segments.

This result is useful precisely because it prevents overclaiming. The segments are valuable as descriptive consumer profiles, but the evidence does not show that they have sharply different overall satisfaction levels.

## 5.2 Concept Performance

The strongest concept pattern is easier to see in observed concept scorecards. Familiar concepts with alcohol service had 374 ratings and a high-satisfaction rate of 46%. Low-price no-alcohol concepts had 294 ratings and a high-satisfaction rate of 33%.

Table 3: Top observed restaurant concept families

| Concept family                         | Ratings | High satisfaction |
|----------------------------------------|---------|-------------------|
| High price — Full bar — Familiar       | 44      | 52.3%             |
| High price — Wine or beer — Familiar   | 121     | 49.6%             |
| Medium price — Wine or beer — Familiar | 142     | 47.9%             |
| Medium price — No alcohol — Familiar   | 387     | 44.4%             |
| Low price — Wine or beer — Quiet       | 17      | 41.2%             |

Top observed restaurant concepts by segment  
 Concepts combine price, alcohol service, and ambience with at least 8 ratings

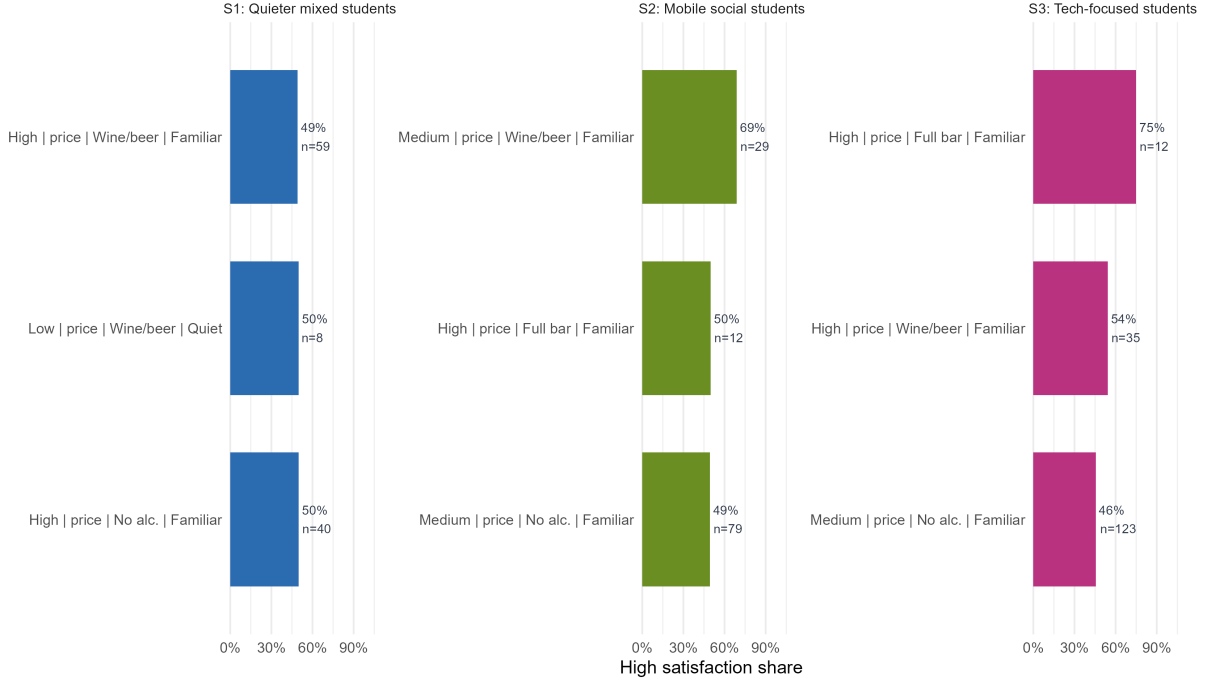


Figure 3: Top observed restaurant concept families within each segment.

H2 was supported in the observed data. The concept-family comparison was statistically significant,  $\chi^2(1) = 12.10$ ,  $p < 0.001$ . Overall, the highest observed concept family was high-price, full-bar, familiar restaurants, with 52% high satisfaction across 44 ratings.

The segment-level scorecard adds useful nuance. Mobile social students rated medium-price, wine/beer, familiar concepts particularly well. Tech-focused students showed a high score for high-price, full-bar, familiar concepts, but that estimate is based on a small cell. A research team should treat that as a hypothesis for a new survey or concept test rather than as a stable market-level number.

### 5.3 Multivariate Driver Analysis

The logistic model tests whether the descriptive concept story survives after controls. The nested-model test supported H3: adding venue attributes improved fit after consumer profile and fit variables were included, likelihood-ratio  $\chi^2(5) = 13.02$ ,  $p = 0.023$ .

### Model-estimated drivers of a high restaurant rating

Odds ratios from a logistic model; intervals are profile-likelihood confidence intervals

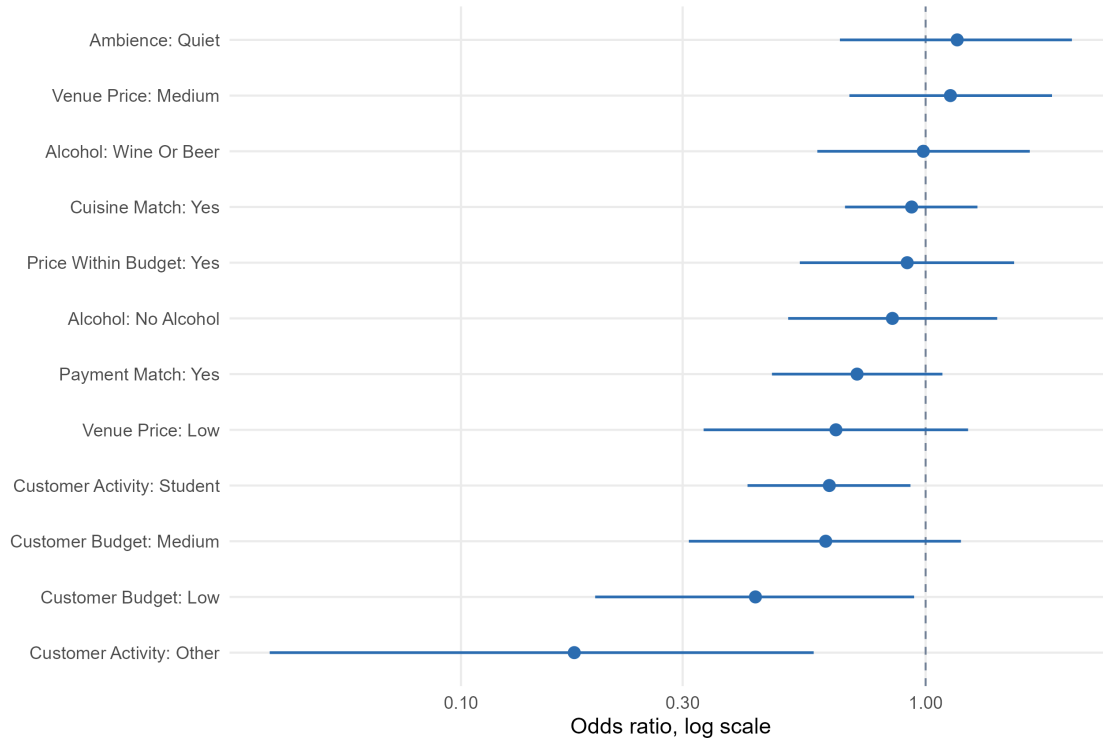


Figure 4: Odds ratios from the high-satisfaction logistic model.

The coefficient-level story is more cautious than the model-comparison result. Low customer budget is associated with lower odds of a high rating relative to the reference group, with an odds ratio of 0.43. Student status is also associated with lower odds relative to the reference activity group, with an odds ratio of 0.62. Individual venue terms such as alcohol service and ambience have wide intervals, so the model does not justify saying that any single venue feature independently causes higher satisfaction.

This is an important difference between a market research signal and a final recommendation. The concept scorecards point toward familiar, higher-service restaurant formats as promising. The regression analysis says that those formats should be tested under a better design before making a causal or investment claim.

Table 4: Hypothesis test summary

| Hypothesis | Test                               | Statistic | p-value |
|------------|------------------------------------|-----------|---------|
| H1         | High rating by segment             | 2.99      | 0.225   |
| H2         | Selected concept-family comparison | 12.10     | < 0.001 |
| H3         | Nested logistic model comparison   | 13.02     | 0.023   |



## 6 Discussion

The strongest answer to the main research question is mixed, but practical. Consumer profile data can be used to create interpretable restaurant consumer segments. Those segments are useful for describing respondents, naming target groups, and building client-facing scorecards. They are not, in this sample, strong predictors of high satisfaction by themselves. This distinction matters because segmentation projects often become too confident once attractive segment names exist. A label such as “mobile social students” is helpful only if the analyst continues to ask what the label predicts and what decision it supports.

The second result is more actionable. Familiar concepts with alcohol service outperform low-price no-alcohol concepts in observed high-satisfaction rates. In a client setting, this would not be enough to recommend changing a restaurant format. It would be enough to prioritize concept cards in a follow-up study. Conjoint analysis and related concept-testing methods are designed for that next step because they expose respondents to controlled attribute combinations rather than relying on whatever combinations happened to appear in the historical data (Green and Srinivasan, 1978, 1990).

The third result helps reconcile the first two. Venue attributes add explanatory value to the model as a group, but the individual feature estimates are not clean. In plain language, restaurant attributes matter, but this data set does not isolate their effects precisely. That conclusion is more credible than forcing a simple story. It shows why descriptive scorecards and multivariate models should be used together: the scorecards are easy to communicate, while the model checks whether the apparent pattern is too dependent on sample composition.

For an Ipsos-style analytics workflow, the final deliverable would be a short set of recommendations:

1. Use the three segments as descriptive planning groups, not as final targeting segments.
2. Treat familiar, higher-service concepts as promising candidates for follow-up testing.
3. Design a primary study with balanced concept exposure, broader recruitment, and explicit purchase-intent or choice outcomes.
4. Keep cell-size thresholds and model uncertainty visible in client materials.

## 7 Limitations

The data set is useful for a market research case study, but it is not a representative restaurant market survey. The sample is small, young, and heavily student-based. That limits generalizability and makes exact segment-level percentages unstable.

The study is observational. Consumers were not randomly assigned to restaurants or concept cards. As a result, the concept comparisons combine respondent preference, restaurant availability, restaurant attributes, and unobserved choice behavior. The logistic model reduces some confounding but does not remove it.

Some concept cells are small even after applying minimum cell-size thresholds. This is especially true for segment-level concept scorecards. The paper therefore treats those cells as screening signals, not as final estimates.

The model is deliberately modest. It uses logistic regression rather than a more complex recommender model because the research question is interpretability and client communication. More

advanced models could improve prediction, but they would not automatically solve the design limitations of the data.

## 8 Conclusion

This analysis supports two practical conclusions. First, mixed-profile clustering can produce interpretable consumer segments from restaurant preference data, but those segments should be validated against outcomes before they are treated as target groups. Second, familiar restaurant concepts with alcohol service show a meaningful observed satisfaction advantage over low-price no-alcohol concepts, and venue attributes improve model fit after consumer controls. The right next step is not to claim that alcohol service or price causes satisfaction. The right next step is to use these findings to design a controlled concept test with a broader respondent base, balanced concept exposure, and explicit behavioral-intent outcomes.

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